Definition of Gradient

Exam code: 9MA0

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Written by: Roger B
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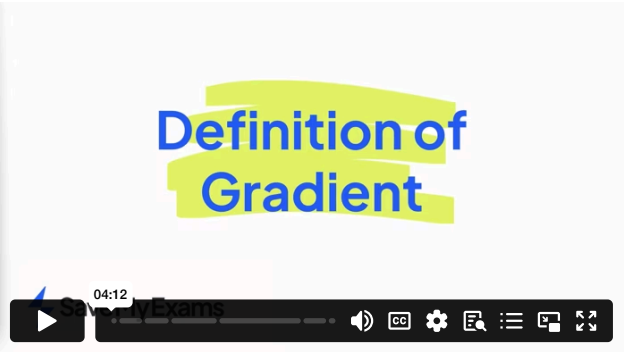

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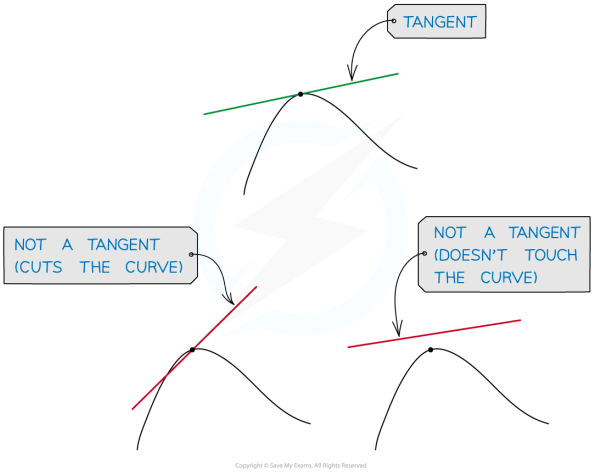
Help me understand

Definition of gradient

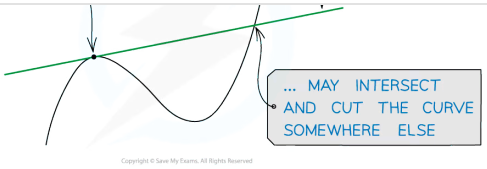
What is the gradient of a curve?

- At a given point the **gradient** of a curve is defined as the gradient of the **tangent** to the curve at that point
- A **tangent** to a curve is a line that just **touches** the curve at one point but doesn't cut the curve at that point



- A tangent *may* cut the curve somewhere else on the curve

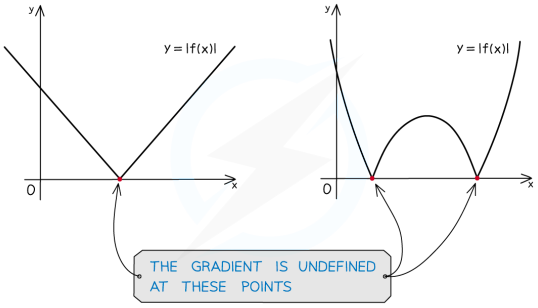
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- It is only possible to draw **one** tangent to a curve at any given point
- Note that unlike the gradient of a straight line, the gradient of a curve is constantly changing

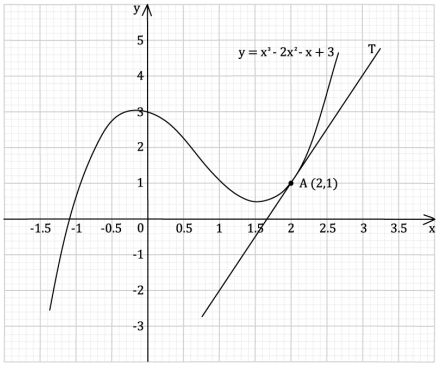
Examiner Tips and Tricks

- A tangent only exists at points where a curve is smooth.
- For example, there is no tangent (or gradient) at one of the 'corners' in a modulus function.

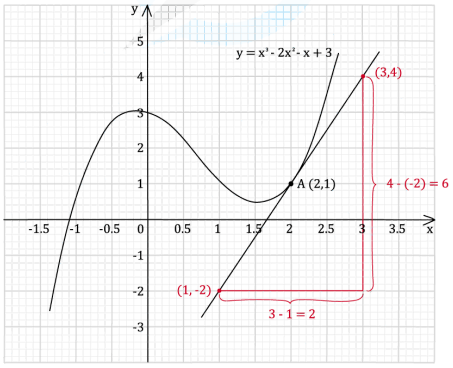


Worked Example

The diagram shows the curve with equation $y = x^3 - 2x^2 - x + 3$. The tangent, T, to the curve at point $A(2, 1)$ is also shown.



Using the diagram, calculate the gradient of the curve at point A.



THE GRADIENT OF THE CURVE AT POINT A IS THE SAME AS THE GRADIENT OF THE TANGENT LINE.

SO

$$\text{GRADIENT} = \frac{\text{CHANGE IN } y}{\text{CHANGE IN } x} = \frac{4-(-2)}{3-1} = \frac{6}{2} = 3$$

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